

```
========================================
        CONTACT MANAGEMENT SYSTEM
========================================
 1. Add contact
 2. Delete contact
 3. Search contact
 4. Update phone number
 5. Display all contacts (forward)
 6. Display all contacts (backward)
 7. Display contacts sorted by name
 8. Show AVL tree traversals
 9. Make a call
10. Add contact to call queue
11. Process next call in queue
12. Show call queue
13. Show recent calls
14. Undo last delete
15. Show deleted contacts history
16. Add connection between contacts
17. Show connections (list and matrix)
18. BFS / DFS from a contact
19. Shortest introduction chain
 0. Exit
Enter your choice: 1

--- ADD NEW CONTACT ---
Enter Contact ID (1-30): 1
Enter Name: Ram
Enter 10-digit Phone Number: 9527338580
Enter Group (Family/Friend/Work): friend
Contact saved successfully.

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18. BFS / DFS from a contact
19. Shortest introduction chain
 0. Exit
Enter your choice: 1

--- ADD NEW CONTACT ---
Enter Contact ID (1-30): 2
Enter Name: sarthak
Enter 10-digit Phone Number: 9356994680
Enter Group (Family/Friend/Work): friend
Contact saved successfully.

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12. Show call queue
13. Show recent calls
14. Undo last delete
15. Show deleted contacts history
16. Add connection between contacts
17. Show connections (list and matrix)
18. BFS / DFS from a contact
19. Shortest introduction chain
 0. Exit
Enter your choice: 2

--- DELETE CONTACT ---
Enter Contact ID to delete: 2
Contact sarthak deleted.

========================================
        CONTACT MANAGEMENT SYSTEM
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 1. Add contact
 2. Delete contact
 3. Search contact
 4. Update phone number
 5. Display all contacts (forward)
 6. Display all contacts (backward)
```



```
Enter your choice: 2

--- DELETE CONTACT ---
Enter Contact ID to delete: 2
Contact sarthak deleted.

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 9. Make a call
10. Add contact to call queue
11. Process next call in queue
12. Show call queue
13. Show recent calls
14. Undo last delete
15. Show deleted contacts history
16. Add connection between contacts
17. Show connections (list and matrix)
18. BFS / DFS from a contact
19. Shortest introduction chain
 0. Exit
Enter your choice: 3

--- SEARCH CONTACT ---
1. Search by ID (linked list)
2. Search by Name (AVL tree)
Enter choice:

...Program finished with exit code 9
Press ENTER to exit console.
```